Docker Cheat Sheet

Made for the SkyListen project. Plain words, only what you will use.

The 4 words

image	The recipe and ingredients for a box. Downloaded once. Example: apache/airflow.
container	A running box made from an image. Sealed, runs the same everywhere.
volume	A disk that lives outside the box, so data (like the database) survives when the box is removed.
compose	One file (docker-compose.yaml) that lists many boxes and starts them together.

Look around (safe, changes nothing)

docker --version	Is Docker installed?
docker compose version	Is Compose installed?
docker ps	List boxes that are running now.
docker ps -a	List all boxes, including stopped ones.
docker images	List downloaded images.
docker system df	How much disk Docker uses.

Run a single box

docker run hello-world	Download and run a tiny test box.
docker logs <name>	Show what a box printed. Add -f to follow live.
docker exec -it <name> bash	Open a terminal inside a running box.
docker stop <name>	Stop a box.
docker start <name>	Start a stopped box again.
docker rm <name>	Delete a stopped box.
docker pull <image>	Download an image without running it.

Compose: run many boxes (run inside the folder with the file)

`docker compose config -q`	Check the file is valid. Prints nothing if fine.
`docker compose up airflow-init`	Run only the one-time setup box.
`docker compose up -d`	Start everything in the background (-d = detached).
`docker compose up`	Start everything and show logs. Ctrl+C stops it.
`docker compose ps`	Which compose boxes are running?
`docker compose logs -f <service>`	Follow one service's logs. Example: airflow-scheduler.
`docker compose restart <service>`	Restart one service.
`docker compose down`	Stop and remove the boxes. Volumes (your data) stay.

Careful: `docker compose down -v` also deletes the volumes, which wipes Airflow's database (its memory of what ran). `docker system prune` deletes unused boxes and images. Both are fine when you mean it, but never run them by habit.

SkyListen routine

`cd ~/Coding/skylisten/airflow`	Compose commands must run in this folder.
`docker compose up airflow-init`	First time only. Sets up the database and login.
`docker compose up -d`	Start Airflow. Web page: http://localhost:8080
`docker compose down`	Stop Airflow when finished for the day.

The collector is **not** in Docker. It runs from the project folder: `caffeinate -i uv run collect.py`